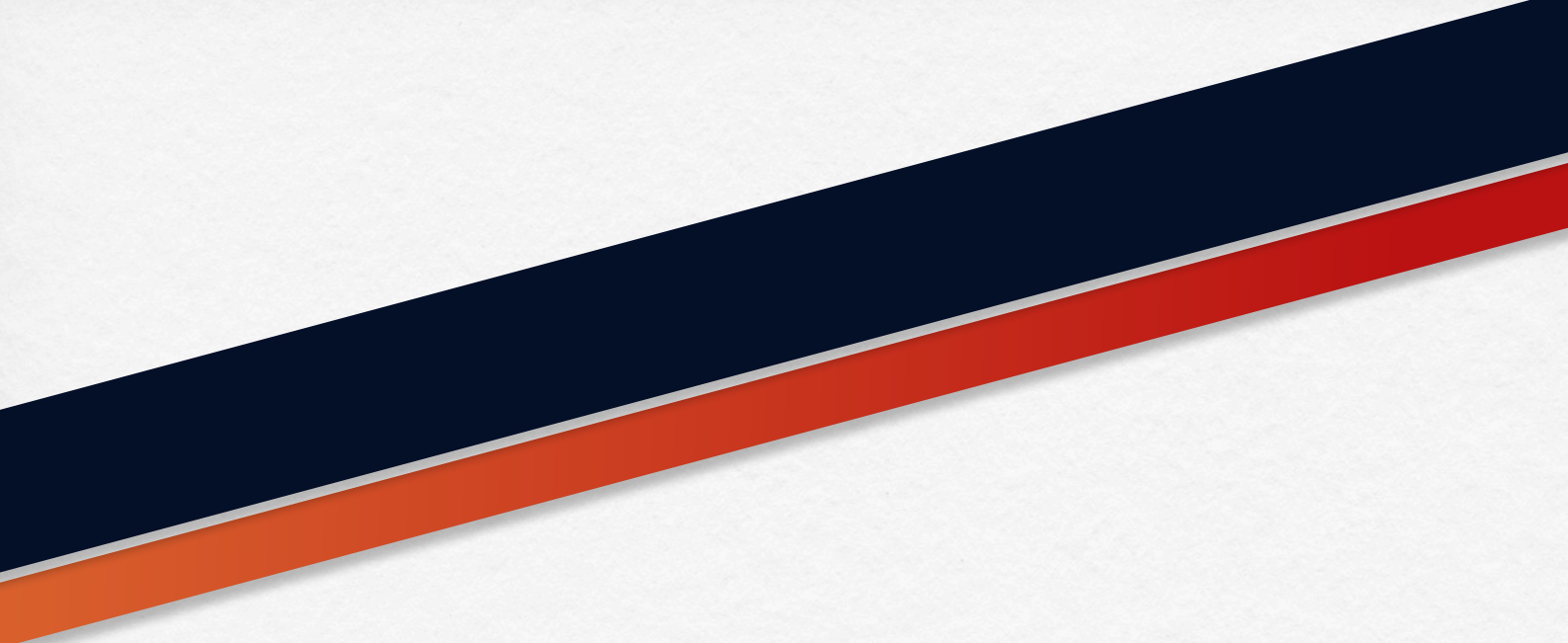




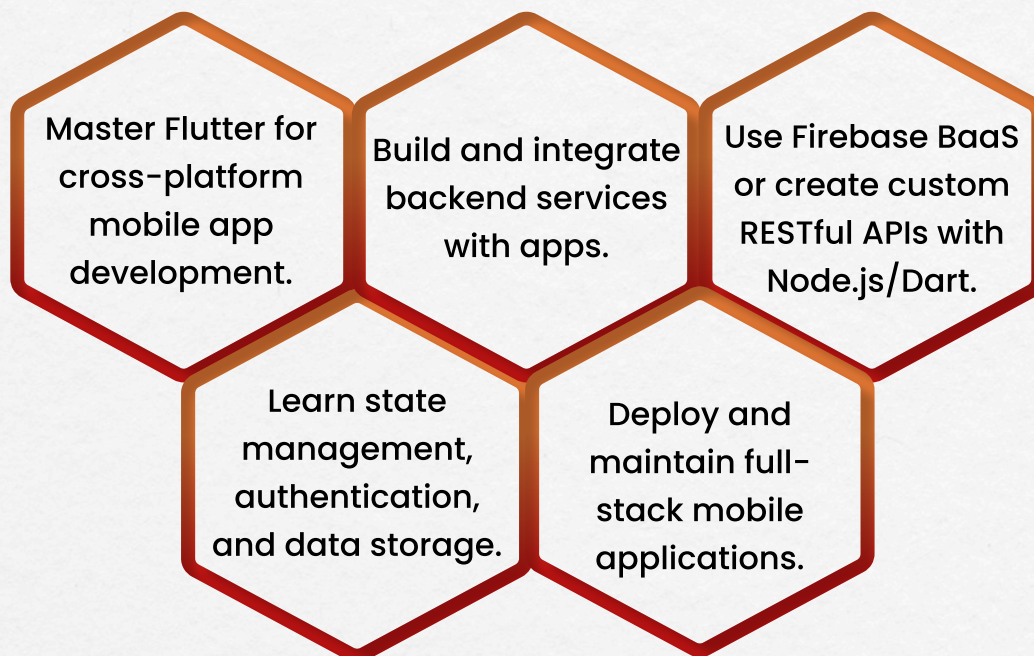
Full Stack Mobile App Development with Flutter (6 months)



Overview

This course provides a comprehensive understanding of full-stack mobile app development using Flutter, Dart, Firebase, and backend technologies. By the end of the course, students will be able to build, deploy, and maintain mobile apps with a robust backend, integrating Flutter for front-end development and Firebase or custom backend solutions for server-side logic.

Course Objectives



Course Outline

1: Introduction to Mobile App Development & Flutter

- Introduction to mobile app development.
- Flutter and Dart basics: Setup, structure, and first Flutter app.
- Widgets, layouts, and navigation.
- Introduction to Dart programming language.
- Development environment setup for Flutter (Android Studio, VSCode).

2: Flutter UI and User Interactions

- Introduction to mobile app development.
- Flutter and Dart basics: Setup, structure, and first Flutter app.
- Widgets, layouts, and navigation.
- Introduction to Dart programming language.
- Development environment setup for Flutter (Android Studio, VSCode).

3: Advanced Flutter UI and Navigation

- Advanced navigation and routing (Named routes, Pass data between screens).
- Animation and transitions in Flutter.
- Working with images, icons, and custom assets.
- Introduction to Flutter packages and libraries.

4: Introduction to Backend Development

- Introduction to RESTful APIs and Web Services.
- Overview of backend technologies: Node.js, Express, Firebase.
- Setting up a basic Node.js server.
- Connecting Flutter frontend with a simple backend (API calls using HTTP).

5: Firebase for Mobile Apps

- Introduction to Firebase (Firestore, Firebase Authentication, Firebase Storage).
- Firebase Authentication for user registration and login.
- Integrating Firestore (cloud database) with Flutter for storing and retrieving data.
- Firebase Storage for handling media files (images, videos).

6: State Management in Flutter

- Overview of state management approaches in Flutter (Provider, Riverpod, Bloc).
- Introduction to scoped model and Provider for managing state.
- Understanding reactive programming in Flutter.
- Using StreamBuilder and FutureBuilder for async data handling.

7: Backend Services with Node.js (or Dart)

- Setting up Node.js and Express for backend development.
- Building RESTful API with Node.js and Express.
- Setting up a database (MongoDB, MySQL) for persistent storage.

8: Mobile App Data Persistence and Offline Functionality

- Local storage options in Flutter (Sqflite, Shared Preferences).
- Working with offline-first design patterns.
- Implementing local data caching and syncing with the cloud.
- Handling app data and persistence with Flutter plugins.

9: Real-Time Communication with Firebase

- Introduction to Firebase Realtime Database.
- Implementing real-time data sync in Flutter apps.
- Building a real-time chat app with Firebase..

10: Testing and Debugging Flutter Apps

- Unit testing and widget testing in Flutter.
- Using the Flutter DevTools for debugging.
- Setting up continuous integration and testing pipelines.
- Best practices for writing testable Flutter code.

11: Deployment and Publishing

- Preparing Flutter apps for production.
- Setting up app icons, splash screens, and app configuration for Android/iOS.
- Deploying to Google Play Store and Apple App Store.
- Versioning and maintenance of mobile apps.

Full-Stack Project

-  Design and develop a full-stack Flutter app with custom backend (Node.js or Firebase).
-  Integrating frontend and backend services.
-  Implementing features like authentication, data storage, and real-time updates.
-  Code reviews and optimization.
-  Final project presentation and feedback.

Final Project:

Build a full-stack mobile application incorporating Flutter for the frontend and Firebase or a custom backend (Node.js, etc.) for the backend. This will be the capstone project for the course.

Evaluation and Grading:

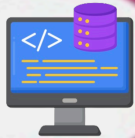
- **Assignments (40%):** Weekly practical assignments.
- **Midterm Project (20%):** A mini-project demonstrating key concepts learned so far.
- **Final Project (30%):** Full-stack mobile app project with frontend and backend integration.
- **Class Participation (10%):** Active participation in class discussions and project presentations.

Tools & Technologies:



Frontend:

Flutter, Dart, Flutter DevTools.



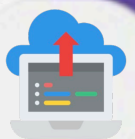
Backend:

Firebase, Node.js, Express, MongoDB, MySQL.



Version Control:

Git, GitHub.



Deployment:

Heroku, Firebase Hosting, Google Play Store, Apple App Store.



Testing:

Flutter test framework, Firebase Emulator.

Unlock the Power of Development: Become A Full Stack Developer with Flutter

Dive into the world of mobile app development and build powerful, cross-platform applications with robust backends. With hands-on projects, expert guidance, and industry-ready tools, this course equips you with the skills to design, develop, and deploy complete mobile solutions.

- ✓ **Real-World Projects –** Build full-stack apps with Flutter, Firebase, and custom APIs.
- ✓ **Top Tools Covered-** Flutter, Dart, Firebase, Node.js, MongoDB, Git & more.
- ✓ **Flexible Learning-** Master app development at your own pace.
- ✓ **Certification-** Industry-recognized credentials to launch your developer career.
- ✓ **Expert Instructors-** Learn from professionals building real-world mobile apps.

**Your Future in
Full-Stack Mobile
App Development
starts here!!!**

Enroll today & start your journey toward becoming a Full-Stack Mobile App Developer..!

"Ready to Build, Deploy & Scale? Join Now."



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